

D3

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n^2}\right)^{n^2} = e$$

D5

$$\lim_{x \rightarrow 0} \frac{\log(1+2x^2)}{\sin(x)} = \frac{\log(1+2x^2)}{2x^2} \cdot 2x^2 \cdot \frac{x}{\sin(x)} \cdot x = 2x^2 \cdot x = 2x^3 = 2 \cdot 0$$

D7

$$y = \frac{x^2 - 1}{x^3}$$

$$y' = \frac{(2x)(x^3) - (x^2 - 1)(3x^2)}{x^6} = \frac{2x^4 - 3x^4 + 3x^2}{x^6} = \frac{x^2(2x^2 - 3x^2 + 3)}{x^6} = \frac{2x^2 - 3x^2 + 3}{x^4} = \frac{-x^2 + 3}{x^4}$$

$$\frac{-x^2 + 3}{x^4} < 0 \Rightarrow -x^2 + 3 < 0 \Rightarrow -x^2 < -3 \Rightarrow x^2 > 3 \Rightarrow x < -\sqrt{3} \vee x > \sqrt{3}$$

D2

$$y = \begin{cases} x^2 & x \leq 0 \\ \sin^2(x) & x > 0 \end{cases}$$

$$f(x) = x^2$$

$$f'(x) = 2x$$

$$f'(x) = 0 \Rightarrow 2x = 0 \Rightarrow x = 0$$

f è crescente per $x > 0$

$$f(x) = (\sin^2(x))$$

$$f'(x) = \cos^2 x \cdot 2x = 2x \cos^2 x$$

$$f'(x) = 0 \Rightarrow 2x \cos^2 x = 0 \Rightarrow 2x = 0 \Rightarrow x = 0$$

f è crescente per $x > 0$

$$\text{MIN} = 0$$

D6

$$f(x) = (x^2 + x + 1)e^x \quad f(0) = (0 + 0 + 1)e^0 = (1)e^0 = 1$$

$$f'(x) = (2x+1)(e^x) + (x^2+x+1)e^x$$

$$2xe^x + e^x + x^2e^x + xe^x + e^x$$

$$(x^2 + 3x + 2)e^x$$

$$f'(0) = (0 + 3 \cdot 0 + 2)e^0 = 2$$

$$f''(x) = (2x+3)(e^x) + (x^2+3x+2)e^x$$

$$2xe^x + 3e^x + x^2e^x + 3xe^x + 2e^x$$

$$(x^2 + 5x + 5)e^x$$

$$f''(0) = (0 + 5 \cdot 0 + 5)e^0 = 5$$

$$1 + 2x + 5 \frac{x^2}{2}$$

D8

$$\int_{-\sqrt{\pi/2}}^{\sqrt{\pi/2}} x \sin(x^2) = \left[\frac{1}{2} \cos(x^2) \right]_{-\sqrt{\pi/2}}^{\sqrt{\pi/2}} = \left(\frac{1}{2} \cos(\pi/2) \right) - \left(\frac{1}{2} \cos(-\pi/2) \right) = 0 - 0 = 0$$

$$\int x \sin(x^2) = \int \frac{x \sin(x^2)}{-2x} = \int \frac{-\sin(y)}{2} = \frac{1}{2} \int -\sin(y) = -\frac{1}{2} \int \sin(y) = \frac{1}{2} \cos y + C = \frac{1}{2} \cos(x^2) + C$$

$$y = x^2$$

$$dy = 2x dx$$

$$\frac{dy}{2x} = dx$$

D4

$$\sum_{n=1}^{\infty} \frac{n+1}{n^2 e} \sim \frac{n}{n^2 e} \quad \text{per convergere } e > 1, \text{ quindi } e$$

Eg 1

$$y = \frac{\ln(x)}{x^2}$$

DOMINIO

$$x > 0 \Rightarrow x \in (0, +\infty)$$

LIMITI

$$\lim_{x \rightarrow +\infty} \frac{\ln(x)}{x^2} = \frac{-\infty}{+\infty} \text{ hospital} = -\infty$$

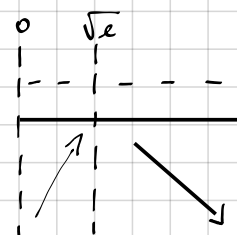
$$\lim_{x \rightarrow 0^+} \frac{\ln(x)}{x^2} = \frac{1}{x} \cdot \frac{1}{2x} = \frac{1}{2x^2} = +\infty$$

MONOTONIA

$$y = \frac{\ln(x)}{x^2}$$

$$y' = \frac{\left(\frac{1}{x}\right)(x^2) - (\ln(x))(2x)}{x^4} = \frac{x - 2x \ln(x)}{x^4} = \frac{x(1 - 2 \ln(x))}{x^4} = \frac{1 - 2 \ln(x)}{x^3}$$

$$\frac{1 - 2 \ln(x)}{x^3} > 0 \quad \left\{ \begin{array}{l} 1 - 2 \ln(x) > 0 \\ x^3 > 0 \end{array} \right. \quad \left\{ \begin{array}{l} -2 \ln(x) > -1 \\ x > 0 \end{array} \right. \quad \left\{ \begin{array}{l} \ln(x) < 1/2 \\ x > 0 \end{array} \right. \quad \left\{ \begin{array}{l} x < e^{1/2} \\ x > 0 \end{array} \right. \quad \left\{ \begin{array}{l} x < \sqrt{e} \\ x > 0 \end{array} \right.$$



↗ crescente per $x \in (0, \sqrt{e})$

↘ decrescente per $x \in (\sqrt{e}, +\infty)$

MAX & MIN

$x = \sqrt{e}$ è max assoluto

FLESSI

$$f'(x) = \frac{1 - 2 \ln(x)}{x^3}$$

$$f''(x) = \frac{\left(-\frac{2}{x}\right)(x^3) - (1 - 2 \ln(x))(3x^2)}{x^6} = \frac{-2x^2 - (3x^2 - 6x^2 \ln(x))}{x^6} = \frac{-2x^2 - 3x^2 + 6x^2 \ln(x)}{x^6} = \frac{x^2(-2 - 3 + 6 \ln(x))}{x^6} = \frac{-5 + 6 \ln(x)}{x^4}$$

$$y'' = \frac{-5 - 6 \ln(x)}{x^4}$$

$$\frac{-5 - 6 \ln(x)}{x^4} = 0$$

$$-5 + 6 \ln(x) = 0$$

$$6 \ln(x) = 5$$

$$\ln(x) = 5/6$$

$$x = e^{5/6}$$

punto di flesso

$$\frac{-5 + 6 \ln(x)}{x^4} > 0 \quad \left\{ \begin{array}{l} -5 + 6 \ln(x) > 0 \Rightarrow 6 \ln(x) > 5 \Rightarrow x > e^{5/6} \\ \forall x \end{array} \right.$$

$f(x)$ convessa per $x \in (e^{5/6}, +\infty)$
 $f(x)$ concava per $x \in (0, e^{5/6})$

2)

$f(x)$ è iniettiva su $[K, +\infty]$ per $K = \sqrt{e}$ in poi

Eg 2

1)

La condizione necessaria per far sì che $\sum a_n$ converga è che $\lim_{n \rightarrow +\infty} a_n = 0$

2)

$$\sum_{n=1}^{\infty} e^{1/n}$$

$$\lim_{n \rightarrow +\infty} e^{1/n} = e^0 = 1$$

$\sum a_n$ non può convergere quindi $\sum$ diverge

Es 3

2)

$$y = x^2 e^{x+1}$$

$$\int x^2 e^{x+1} = x^2 e^{x+1} - \int 2x e^{x+1} = x^2 e^{x+1} - 2 \int x e^{x+1} = x^2 e^{x+1} - 2 \left(x e^{x+1} - \int e^{x+1} \right) = x^2 e^{x+1} - 2x e^{x+1} + 2 e^{x+1}$$

$$e^{x+1} (x^2 - 2x + 2) + C$$

$$\int e^{x+1} = \int e^y = \int e^y = e^{x+1} + C$$

$$y = x+1$$

$$dy = dx$$

1) Una funzione $f(x)$ continua su un intervallo chiuso $[a, b]$ esiste almeno un punto x_0 tale che $f(x_0) = \frac{1}{b-a} \int_a^b f(x)$

3)

$$\left[e^{x+1} (x^2 - 2x + 2) \right]_0^2 = (e^{2+1} (4 - 4 + 2)) - (e^{1+1} (0 - 0 + 2))$$

$$2e^3 - 2e = 2(e^3 - e)$$

$$\frac{1}{2-0} \cdot 2(e^3 - e) = e^3 - e$$